

Partner.skill: A Quantitative Framework for AI-Assisted Relationship Maintenance

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Abstract

This paper introduces **Partner.skill**, an AI-driven relationship maintenance and enhancement engine designed to operate within the OpenClaw and Claude Code environments. Unlike traditional conversational agents that offer generic dating advice, Partner.skill constructs a highly personalized, dynamic model of a romantic partner by analyzing multimodal inputs (chat logs, emails, and behavioral descriptions). The system integrates established psychological frameworks—including Attachment Theory, the Big Five personality traits, and Gottman’s Four Horsemen—with a novel mathematical model, the Relationship Quality Index (RQI). By quantifying relationship health and simulating partner-specific responses across 23 distinct life scenarios, Partner.skill provides actionable, evidence-based strategies for conflict resolution, emotional support, and long-term relationship development.

1. Introduction

The application of artificial intelligence in interpersonal relationships has largely focused on matchmaking or simulating companionship. However, there is a significant gap in tools designed to maintain and enhance *existing* relationships. Partner.skill addresses this gap by functioning as a “meta-skill”—a system that distills a partner’s communication patterns, emotional triggers, and personality traits into a localized, privacy-preserving AI advisor.

The core philosophy of Partner.skill is that effective relationship maintenance requires more than empathy; it requires structural understanding. By translating qualitative interactions into quantitative metrics, the system helps users navigate complex

emotional landscapes, from daily chore negotiations to deep-seated attachment conflicts.

2. Theoretical Foundations

Partner.skill's analytical engine is grounded in several validated psychological and sociological frameworks:

2.1 Attachment Theory

Based on Bowlby's (1969) work, the system categorizes partners into Secure, Anxious, Avoidant, or Fearful-Avoidant styles. This classification fundamentally alters the AI's advice; for instance, conflict resolution strategies for an Avoidant partner emphasize space and logical processing, whereas strategies for an Anxious partner prioritize immediate reassurance [1].

2.2 The Five Love Languages

Chapman's (1992) framework is utilized to align expressions of affection. The system calculates a Love Language Mismatch Index (LLMI) to identify areas where partners may be expressing love in ways the other does not naturally receive [2].

2.3 Gottman's Four Horsemen

The conflict resolution module actively scans input data for Criticism, Contempt, Defensiveness, and Stonewalling. Identifying these behaviors triggers specific, structured repair pathways designed to de-escalate tension [3].

2.4 The Big Five (OCEAN)

Personality traits (Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism) are extracted to predict behavioral tendencies in scenarios such as travel planning or financial discussions [4].

3. System Architecture

The Partner.skill architecture consists of two primary components that operate in tandem:

3.1 Part A: Relationship Profile

This component maintains the state of the relationship. It stores the calculated RQI, historical conflict patterns, and a weighted memory bank. The memory bank utilizes an activation function: $W = E \times R \times (1 + F)$ where E is Emotional Intensity, R is Recency, and F is Frequency. Memories with a weight exceeding a predefined threshold are classified as “Core Memories” and heavily influence the system’s contextual understanding.

3.2 Part B: Persona

The Persona component simulates the partner’s communication style and decision-making processes. It is structured in a hierarchical layer model:

- **Layer 0 (Hard Rules):** Explicit user-defined constraints (e.g., “Never apologizes first”).
- **Layer 1 (Identity):** Baseline psychological traits (MBTI, Attachment Style).
- **Layer 2 (Expression):** Syntactic habits, emoji usage, and vocabulary extracted from chat logs.
- **Layer 3 (Decision Making):** Cognitive processing styles under stress.
- **Layer 4 (Correction):** Iterative user feedback that overrides lower layers.

4. The Relationship Quality Index (RQI) Model

The cornerstone of Partner.skill is the RQI, a proprietary mathematical model that quantifies relationship health.

4.1 RQI Calculation

The base RQI is calculated as a weighted sum of five dimensions: $\text{Base RQI} = \sum(w_i \times s_i)$

Dimension	Weight (w_i)	Description
Communication Patterns	0.30	Frequency of active listening vs. hostile exchanges
Emotional Intimacy	0.25	Depth of vulnerability and mutual support
Love Language Alignment	0.20	Compatibility of primary affection expressions
Conflict Repair Capacity	0.15	Speed and effectiveness of post-argument recovery
Growth Trajectory	0.10	Shared goals and individual support

4.2 Attachment Compatibility Score (ACS)

The Base RQI is then modified by the ACS, a multiplier derived from the interaction of both partners' attachment styles. $\text{Final RQI} = \text{Base RQI} \times \text{ACS_modifier}$

For example, a Secure-Secure pairing yields an ACS of 0.95, amplifying the Base RQI, while an Anxious-Avoidant pairing yields an ACS of 0.30, reflecting the inherent structural challenges of the dynamic.

5. Scenario-Based Application

Partner.skill operationalizes its theoretical models across 23 distinct life scenarios, categorized into four domains:

- Emotional & Conflict:** e.g., handling jealousy, sincere apologies.
- Celebration & Gifting:** e.g., anniversary planning based on Love Languages.
- Experience & Daily:** e.g., date planning tailored to Big Five traits.
- Practical Life:** e.g., chore negotiation, financial discussions.

When a user queries a scenario, the system first evaluates the Persona (Part B) to predict the partner's likely reaction, then consults the Relationship Profile (Part A) to formulate a strategy that maximizes connection and minimizes friction.

6. Privacy and Ethical Considerations

Given the highly sensitive nature of relationship data, Partner.skill is designed as a local-first application. All data parsing, profile generation, and scenario analysis occur entirely within the user's local environment (Claude Code or OpenClaw). No chat logs or psychological profiles are transmitted to external servers. Furthermore, the system includes ethical guardrails to flag signs of emotional abuse or coercive control, advising professional intervention when necessary.

7. Conclusion

Partner.skill represents a significant advancement in personal AI applications. By moving beyond generic advice and employing rigorous, quantitative models of personality and relationship dynamics, it offers users a powerful tool for understanding and nurturing their romantic partnerships. Future iterations will focus on expanding the memory activation model and integrating real-time physiological data (e.g., heart rate variability during conflicts) for even more precise emotional mapping.

References

[1] Bowlby, J. (1969). *Attachment and loss: Vol. 1. Attachment*. Basic Books. [2] Chapman, G. (1992). *The Five Love Languages: How to Express Heartfelt Commitment to Your Mate*. Northfield Publishing. [3] Gottman, J. M., & Silver, N. (1999). *The Seven Principles for Making Marriage Work*. Crown. [4] McCrae, R. R., & Costa, P. T. (1987). Validation of the five-factor model of personality across instruments and observers. *Journal of Personality and Social Psychology*, 52(1), 81-90.